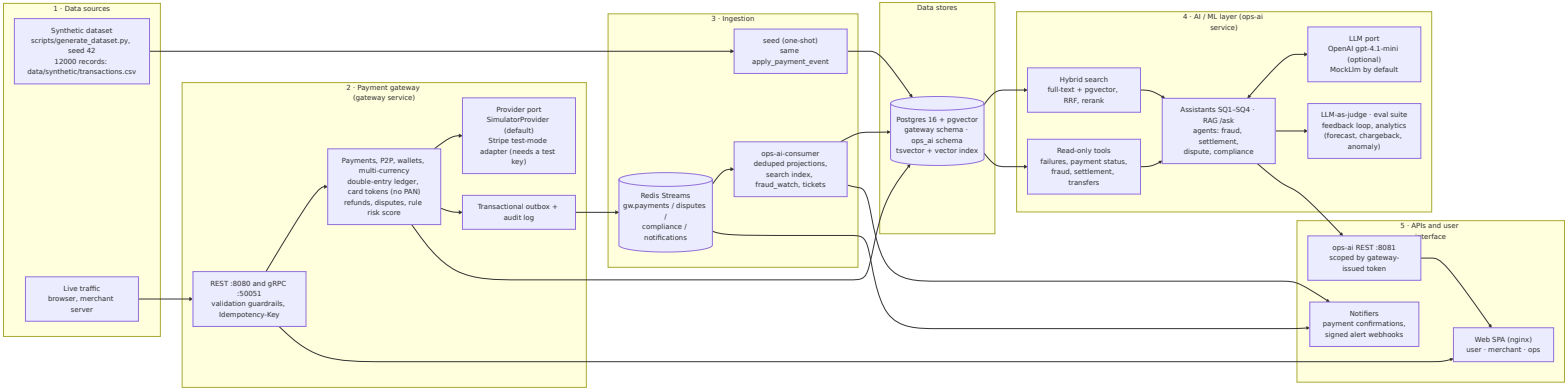
1 · Data sources
Synthetic dataset
scripts/generate_dataset.py,
seed 42
12000 records:
data/synthetic/transactions.csv
Live traffic
browser, merchant
server
2 · Payment gateway
(gateway service)
REST :8080 and gRPC
:50051
validation guardrails,
Idempotency-Key
Payments, P2P, wallets,
multi-currency
double-entry ledger,
card tokens (no PAN)
refunds, disputes, rule
risk score
Provider port
SimulatorProvider
(default)
Stripe test-mode
adapter (needs a test
key)
Transactional outbox +
audit log
3 · Ingestion
seed (one-shot)
same
apply_payment_event
Redis Streams
gw.payments / disputes
/
compliance /
notifications
ops-ai-consumer
deduped projections,
search index,
fraud_watch, tickets
Data stores
Postgres 16 + pgvector
gateway schema ·
ops_ai schema
tsvector + vector index
4 · AI / ML layer (ops-ai
service)
Hybrid search
full-text + pgvector,
RRF, rerank
Read-only tools
failures, payment status,
fraud, settlement,
transfers
Assistants SQ1–SQ4 ·
RAG /ask
agents: fraud,
settlement,
dispute, compliance
LLM port
OpenAI gpt-4.1-mini
(optional)
MockLlm by default
LLM-as-judge · eval suite
feedback loop, analytics
(forecast, chargeback,
anomaly)
5 · APIs and user
interface
ops-ai REST :8081
scoped by gateway-
issued token
Notifiers
payment confirmations,
signed alert webhooks
Web SPA (nginx)
user · merchant · ops